

Shayan Hussain

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PROFESSIONAL SUMMARY

Final-year Computer & Information Systems Engineering student with a proven track record of deploying production-grade RAG pipelines, LLM integrations, and MLOps workflows. Seeking a remote-first Applied AI role to drive scalable, real-world automation through agentic systems and full-stack development.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, Dart, HTML, CSS, SQL

AI & Machine Learning: RAG Pipelines, LLMs, LangChain, Vector Search, MLOps, TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLO, CNNs, NLP, Transformers, Reinforcement Learning, Pandas, NumPy

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Containerization, Nginx, GitHub Actions (CI/CD), GHCR

Frameworks & Tools: FastAPI, Next.js (App Router), React, Node.js, Celery, Drizzle ORM, TailwindCSS, shadcn/ui, Django, Flask, MERN Stack, Flutter, MCP, n8n, Git

Databases & Storage: PostgreSQL, pgvector, Redis (Semantic Caching), MySQL, Supabase, SQLite

Core Competencies: Agentic AI, Multi-Tenancy Architecture, Production Deployment, Computer Vision, Full-Stack AI SaaS

WORK EXPERIENCE

Shispare

AI Engineer (Full-time)

Karachi, Pakistan

January 2026 – May 2026

- Designed and deployed a production-grade RAG system for the Stellar LMS platform, successfully automating the generation of assessment questions from PDF course content in a live environment.
- Engineered an operational Model Context Protocol (MCP) server for Pipedrive CRM within a 4-person team, automating workflows to eliminate manual effort by 50% and achieving single-digit second response times for AI agent integrations.
- Architected an agentic coding assistant via a VS Code extension, utilizing locally deployed LLMs and a centralized RAG index over Bitbucket/GitHub to ensure enterprise data privacy while accelerating developer workflows.
- Optimized and deployed local LLMs, reducing inference and response times by up to 30% to decrease API dependency and enhance internal tooling efficiency.

10Pearls

MERN Stack Intern

Remote

September 2025 – November 2025

- Built a scalable, production-ready Notes Application utilizing the MERN stack with robust CRUD functionality, achieving 90%+ test coverage through rigorous Jest unit testing.
- Partnered directly with a senior engineer to execute asynchronous development cycles, managing modular feature branches via GitHub to support seamless remote delivery.

Softech Digital Group

Software Development Intern

Hybrid

September 2025 – November 2025

- Contributed to the development of a React and FastAPI-based Art DRM Marketplace, successfully integrating PayPal to process secure payment gateways.
- Prototyped AI-based virtual Try-On technology integration approaches, validating technical feasibility for commercial product fitting use cases.

TECHNICAL PROJECTS

Deflekt — Multi-Tenant AI Support Deflection SaaS

2026

AWS EC2/RDS, Next.js 14, FastAPI, pgvector, Redis, Celery, Docker | [Live Demo](#) | [GitHub](#)

- Architected a multi-tenant RAG platform using Next.js 14, Python FastAPI, PostgreSQL (pgvector), and Redis, reducing repetitive support ticket volume by 70% with cited, grounded answers.
- Engineered a sub-18ms Redis semantic cache cutting duplicate LLM query latency by 98.5% and scaled single-instance AWS EC2 throughput to 150+ req/sec (cached) and 45+ req/sec (streaming RAG), supporting 10,000+ daily active users.
- Deployed an async Celery worker queue processing 1,200 words/sec for document ingestion and built an automated CI/CD pipeline on AWS EC2/RDS with an LLM-as-a-Judge eval harness gating builds on >80% Faithfulness precision.
- Enforced strict database-level multi-tenancy isolation and an automated confidence gate (<0.72) that escalates low-confidence queries to human support with <0.8% hallucination rate.

RoboComm: Smart Interactive Campus Robot

2025 – Present

- Built an autonomous campus assistant using a custom ESP32 chassis integrated with an MPU6050 (heading estimation) and HC-SR04 (obstacle detection), achieving a navigation accuracy of 85%.
- Deployed real-time YOLO object detection maintaining 25–30 FPS and integrated the Gemini API for natural language STT/TTS interaction, enabling voice-driven campus navigation.

Voice-Enabled Booking Automation System

2026

- Architected a zero-touch conversational agent utilizing Make.com workflows to orchestrate Calendly, Google Calendar, and OpenAI APIs, streamlining the booking lifecycle to a sub-5-minute process and successfully processing 100+ automated user interactions.

Anomaly Detector Computer Vision Pipeline

2026

- Developed an end-to-end MLOps pipeline (hand-annotating 1,500 images via CVAT and augmenting via Roboflow) to train an RT-DETR-L model with a frozen backbone, achieving 88% accuracy and a peak mAP of 0.650 for anomaly detection on unseen datasets.

Remote Work Productivity Tracker

2026

- Trained a behavioral classification model on a custom, annotated screen-recording dataset of 1,000 hours to track verified work hours with 80% accuracy, building a real-time Next.js/Node.js analytics dashboard for administrative oversight.

EDUCATION

NED University of Engineering and Technology

Karachi, Pakistan

Bachelor of Engineering in Computer and Information Systems Engineering 2022 – 2026 (Expected)

CGPA: 3.556 / 4.0 | Latest SGPA: 3.95 / 4.0